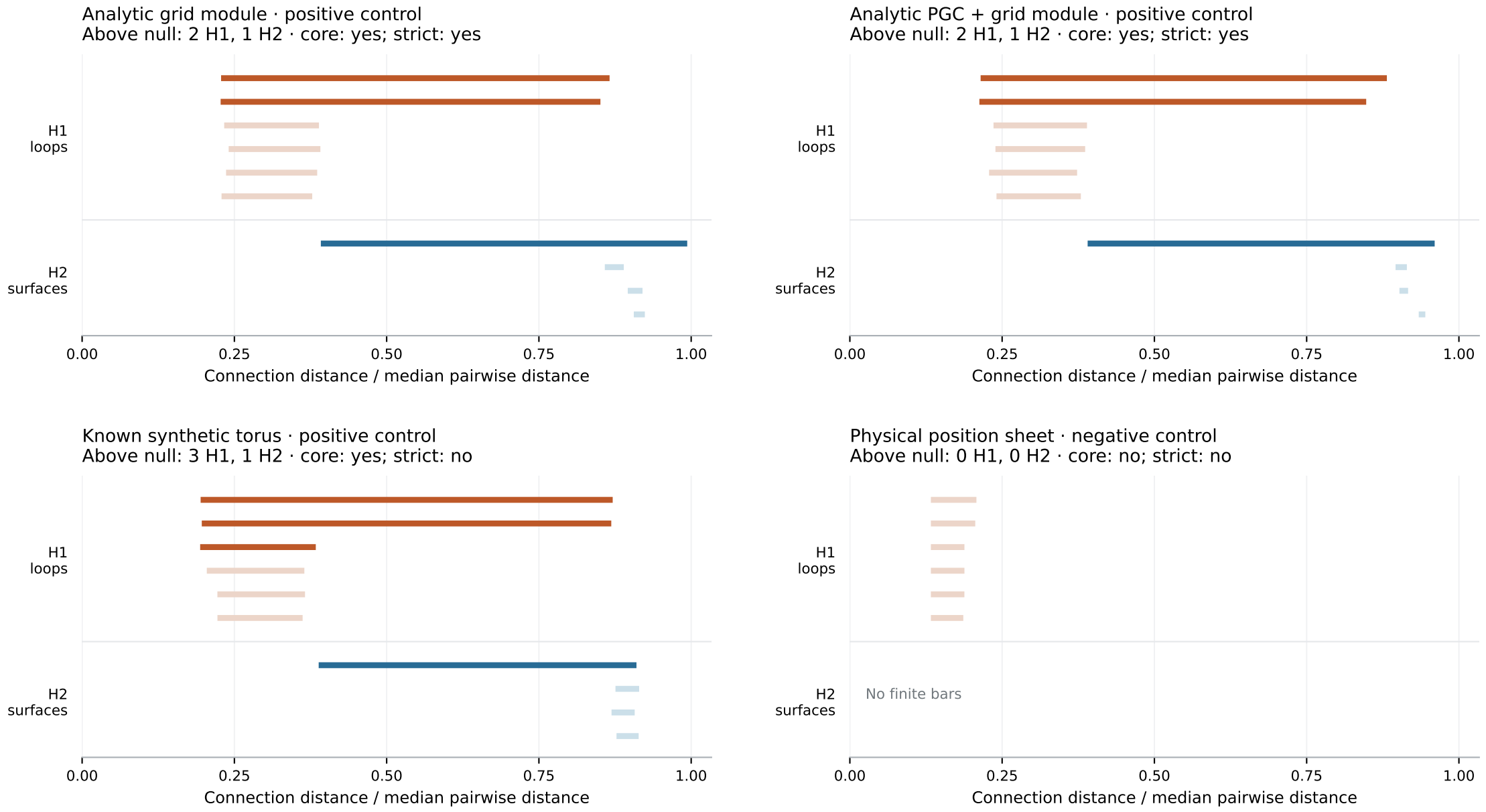
Reference shapes · can this analysis detect a torus?
256 landmarks · original activity distances · no UMAP



Grid-module controls use the same spatial samples as the LSTMs. A clear torus has two dominant H1 loops and one H2 surface.
Dark bars exceed their dimension-specific shuffle threshold; pale bars do not. Top six H1 and four H2 bars shown.
The synthetic torus has an extra short loop that ends before its torus interval; core passes, strict fails.
These are distance-scale barcodes, not temporal trajectories; the physical position sheet is not expected to be toroidal.